Mathematical Models in Biotechnology

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Abstract

Intracellular fluxes, kinetic parameters, and metabolite concentrations are sparsely measured at genome scale. Flux balance analysis estimates fluxes from network stoichiometry, reaction bounds, and an optimization objective. This chapter derives the flux constraint from a mole balance and states the assumptions that lead to its steady-state form. It then writes each reaction bound in terms of reversibility, enzyme capacity, substrate saturation, enzyme abundance, and regulation. A urea-cycle model shows how public thermodynamic and kinetic estimates set reaction bounds. A feedback-inhibited pathway shows how expression and allosteric regulation change a capacity bound. Two larger studies illustrate related methods in cell-free protein synthesis and strain design. Together, these examples show how condition-specific data and proposed interventions change flux predictions through reaction bounds.

1 Introduction

Biotechnology depends on predicting and directing cellular metabolism. Metabolic engineering redirects flux toward a product. Bioprocess development selects operating conditions that sustain production. Strain design identifies genetic changes that alter the phenotype. Each task requires an estimate of metabolic reaction rates under a specified condition [1]. These rates are difficult to measure directly. Genome sequences and reaction annotations provide a network structure, but reconstructions remain uncertain in reaction membership, directionality, compartments, cofactors, and biomass composition. Fluxes, kinetic parameters, and metabolite concentrations are even less complete at genome scale. Models must therefore estimate phenotype from incomplete measurements. Bioprocess models use several levels of biological detail. Unstructured models track biomass, extracellular substrates, and products with lumped growth, uptake, and production rates. Monod growth, inhibition, maintenance, and growth-associated production laws support batch, fed-batch, and continuous reactor analysis, but do not resolve intracellular fluxes [2]. Structured kinetic models add intracellular pools, enzymes, or cellular machinery. Early single-cell models from Shuler and coworkers coupled intracellular metabolism to extracellular nutrients in microbial and mammalian cells [3, 4]. Reaction-level kinetic models assign mass-action, Michaelis–Menten, or power-law rates to individual reactions and integrate the resulting differential equations [5–8]. These models describe transient intracellular behavior, but require more kinetic parameters as their resolution increases.

Constraint-based models use a different closure. They impose mass conservation on the reaction network and require fewer explicit kinetic parameters. This approach can be applied to genome-scale reconstructions [1, 9]. Flux balance analysis (FBA) formulates the calculation as a linear program [10, 11]. Conservation alone usually permits many flux distributions. Reaction bounds restrict this feasible set, and an objective selects an optimum. Bounds are a major route by which condition-specific thermodynamic, kinetic, expression, and regulatory data enter the model.

This chapter examines how each type of data changes a bound and its predicted fluxes. The chapter first derives the flux constraint from a mole balance, including the growth-dilution term $\mathbf{S}\hat{\mathbf{v}} = \mu\mathbf{x}$. It then factors each reaction bound into terms for reversibility, enzyme capacity, substrate saturation, enzyme abundance, and regulation. Two small examples show how these terms affect a flux prediction. The first uses public data to set bounds in a urea-cycle network. The second transfers expression and metabolite states from a kinetic feedback model into an FBA bound. Two larger studies illustrate related methods in dynamic cell-free protein synthesis [12] and model-guided strain design [13]. Strain design also requires a discrete search over candidate interventions, often formulated as a bilevel optimization problem [14].

2 From Mole Balances to the Flux Constraint

The FBA constraint follows from a mole balance. Its derivation defines the normalization basis and identifies the steady-state assumptions. Consider a well-mixed culture with biomass concentration B and reactor volume $\bar{V}$. Their product defines the biomass basis

$$\mathcal{B} \equiv B\bar{V}, \tag{1}$$

where B has units gDW/L, $\bar{V}$ has units L, and $\mathcal{B}$ is the total biomass in gDW. An intracellular concentration C_i has units mmol/gDW, and an intracellular flux $\hat{v}_j$ has units mmol/gDW/h. Multiplication by $\mathcal{B}$ gives the total intracellular amount $C_i\mathcal{B}$ and total reaction rate $\hat{v}_j\mathcal{B}$. A general intracellular balance includes reaction, active transport, and physical transport. Let $c_{i,s}$ be the concentration of species i in a hypothetical bulk stream s, and let $\dot{V}_s$ be its volumetric flow rate. The intracellular mole balance is

$$\sum_s d_s c_{i,s}\dot{V}_s + \mathcal{B}\sum_j \sigma_{ij}\hat{v}_j = \frac{d}{dt}\big(C_i\mathcal{B}\big), \tag{2}$$

where $d_s = +1$ for an inlet and $d_s = -1$ for an outlet, and σ_{ij} is entry (i,j) of the stoichiometric matrix $\mathbf{S} \in \mathbb{R}^{m\times n}$. Each term has units mmol/h. The stream concentration $c_{i,s}$ has units mmol/L, whereas C_i has units mmol/gDW. The cell membrane excludes bulk physical flow from the intracellular control volume, so $\dot{V}_s = 0$ and the first term vanishes. Active transport remains in $\hat{\mathbf{v}}$ because it moves molecules without bulk volume flow.

Expand the accumulation term before assuming a fixed reactor volume. Substitution of $\mathcal{B} = B\bar{V}$ gives

$$\frac{d}{dt}\big(C_i\mathcal{B}\big) = B\bar{V}\frac{dC_i}{dt} + C_i\bar{V}\frac{dB}{dt} + C_iB\frac{d\bar{V}}{dt}. \tag{3}$$

After division by $\mathcal{B} = B\bar{V}$, Equation (2) becomes

$$\frac{1}{\mathcal{B}}\sum_s d_s c_{i,s}\dot{V}_s + \sum_j \sigma_{ij}\hat{v}_j = \frac{dC_i}{dt} + \frac{C_i}{B}\frac{dB}{dt} + \frac{C_i}{\bar{V}}\frac{d\bar{V}}{dt}. \tag{4}$$

For a batch or fed-batch culture with no biomass flow, total biomass changes through growth. The specific growth rate is

$$\mu \equiv \frac{1}{\mathcal{B}}\frac{d\mathcal{B}}{dt} = \frac{1}{B}\frac{dB}{dt} + \frac{1}{\bar{V}}\frac{d\bar{V}}{dt}. \tag{5}$$

The membrane removes the physical-flow term from Equation (4). Balanced growth also gives $dC_i/dt = 0$. Equation (5) then reduces the balance to

$$C_i\mu = \sum_j \sigma_{ij}\hat{v}_j \qquad \Longleftrightarrow \qquad \boxed{\mathbf{S}\hat{\mathbf{v}} = \mu\mathbf{x}}, \tag{6}$$

where $\mathbf{x} = (C_1, \ldots, C_m)^\top$ contains the intracellular concentrations in mmol/gDW and $\hat{\mathbf{v}} \in \mathbb{R}^n$ contains the fluxes in mmol/gDW/h. Net production balances growth dilution. For a fixed-volume batch reactor, $d\bar{V}/dt = 0$ and $\mu = (1/B)dB/dt$. For fed-batch operation, $d\bar{V}/dt \neq 0$, so the volume term must remain. The fixed-volume result is a special case of the general balance.

Conventional FBA neglects the dilution term and uses

$$\mathbf{S}\hat{\mathbf{v}} = \mathbf{0}. \tag{7}$$

This approximation is appropriate when $\mu\mathbf{x}$ is small relative to the metabolic fluxes [10]. Setting $\mu = 0$ is not sufficient. A cell-free batch has no growing biomass, but its transcript, protein, and metabolite concentrations may still change. Its accumulation term $d\mathbf{x}/dt$ must remain. All model classes introduced above use the same balance but close the reaction term differently. Unstructured models use lumped rates, structured kinetic models use state-dependent intracellular rates, and constraint-based models use bounds and an objective. Constraint-based reconstructions represent uptake and secretion with transport and exchange reactions. A boundary reaction $\emptyset \rightleftharpoons M_i$ enters $\mathbf{S}$ as an isolated nonzero column with no intracellular reaction partner [1]. Its flux describes molecular transport across the modeled boundary, not a reactor stream. The included boundary reactions and their bounds define the modeled system boundary.

3 The Linear Program and Its Geometry

The balance equations conserve mass but usually admit many flux distributions; flux bounds restrict this set to a feasible polytope, and an objective selects an optimal vertex or face (Fig. 1). FBA adds lower and upper bounds, $\boldsymbol{\ell} \leq \hat{\mathbf{v}} \leq \mathbf{u}$, and a linear objective $\mathbf{c}^\top\hat{\mathbf{v}}$. Bounds can represent reaction direction, enzyme capacity, and substrate availability. The objective states the quantity to maximize. These elements define the linear program

$$\begin{aligned} \max_{\hat{\mathbf{v}} \in \mathbb{R}^n} \quad & \mathbf{c}^\top\hat{\mathbf{v}} \\ \text{subject to} \quad & \mathbf{S}\hat{\mathbf{v}} = \mathbf{0} \;\; (\text{or } \mu\mathbf{x}), \\ & \boldsymbol{\ell} \leq \hat{\mathbf{v}} \leq \mathbf{u}. \end{aligned} \tag{8}$$

This is the standard FBA formulation [10]. The objective is a modeling choice. Setting $c_j = 1$ for a biomass pseudo-reaction maximizes growth. The biomass reaction consumes precursors in the proportions needed to form new cell mass. Setting $c_j = 1$ for a product-export reaction instead maximizes product secretion. The constraints remain the same; only $\mathbf{c}$ changes. The growth-aware form remains linear when $\mathbf{x}$ is measured and μ is fixed or enters as a scalar decision variable. If both μ and $\mathbf{x}$ are unknown, their product is bilinear and the problem is no longer an LP. A model that retains $\mu\mathbf{x}$ must also define its relation to the biomass pseudo-reaction so that growth-associated demands are not counted twice. The conservation equations usually admit many flux vectors. The null space

$$\mathcal{N}(\mathbf{S}) = \{\hat{\mathbf{v}} \in \mathbb{R}^n : \mathbf{S}\hat{\mathbf{v}} = \mathbf{0}\}$$

contains every steady-state flux vector. If $\operatorname{rank}(\mathbf{S}) < n$, the null space contains more than one vector. Bounds restrict it to a feasible polytope. The objective selects a vertex or a face of that polytope. A face contains multiple optimal flux vectors, so optimization does not always give a unique solution.

The singular value decomposition identifies both null spaces. Write $\mathbf{S} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$. Columns of $\mathbf{V}$ associated with zero singular values span the right null space, so every steady-state flux vector is

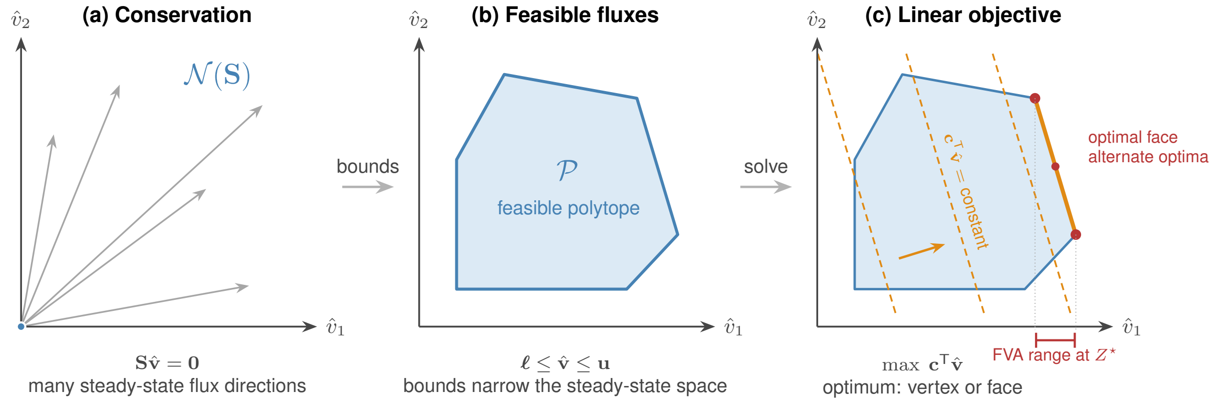

Figure 1: Geometry of FBA in a two-flux projection. (a) Conservation restricts steady-state fluxes to the null space $\mathcal{N}(\mathbf{S})$. (b) Lower and upper bounds form the feasible polytope $\mathcal{P}$. (c) The objective moves toward larger values until it reaches an optimal vertex or face. Fluxes can vary along an optimal face without changing the objective. FVA reports the resulting interval for each reaction, shown here for $\hat{v}_1$.

a linear combination of these columns. Columns of $\mathbf{U}$ associated with zero singular values span the left null space. A vector $\mathbf{y}$ in the left null space defines a conserved quantity $\mathbf{y}^\top \mathbf{x}$, such as the total concentration of a cofactor pair. Genome-scale models often have a large right null space because $\text{rank}(\mathbf{S}) < n$ [1]. Flux variability analysis (FVA) measures the remaining freedom in each reaction [11]. FVA fixes the objective at its optimum, or at a chosen fraction of that optimum, and then minimizes and maximizes each flux. Equal minimum and maximum values mean that a reaction is fixed on that optimal face. A wide interval means that the reaction can vary without changing the required objective value. A different objective can select a different face and produce different intervals.

4 Integrative Flux Bounds

Reaction bounds can include several types of condition-specific data. For reaction j, write the bound as

$$-\delta_j \Big[V^\circ_{\max,j}\,(e/e^\circ)\,\theta_j(\cdot)\,f_j(\cdot) \Big] \;\le\; \hat{v}_j \;\le\; V^\circ_{\max,j}\,(e/e^\circ)\,\theta_j(\cdot)\,f_j(\cdot), \tag{9}$$

where $V^\circ_{\max,j}$ is a reference capacity with units of flux. The other factors are dimensionless. The reversibility factor δ_j sets the reverse direction. The terms $V^\circ_{\max,j}$ and f_j describe enzyme capacity and substrate saturation. The ratio (e/e°) describes enzyme abundance, and θ_j describes allosteric activity. Transcriptional and translational controls u_j and w_j act through the expression model and change (e/e°). They are distinct from θ_j. When data are unavailable, the corresponding factor is set to one. Measurements or mechanistic models can then replace that default.

The reversibility factor δ_j has two values:

$$\delta_j \;=\; \begin{cases} 1, & \text{reaction } j \text{ reversible,} \\ 0, & \text{reaction } j \text{ irreversible,} \end{cases} \tag{10}$$

When $\delta_j = 1$, the lower bound is negative and reverse flux is allowed. When $\delta_j = 0$, the lower bound is zero. A simple assignment compares ΔG°_j with a threshold ΔG^*_j. This threshold rule is heuristic. A thermodynamic analysis should instead propagate stated activity ranges through $\Delta G = \Delta G^\circ + RT\ln Q$ and test whether ΔG can change sign. eQuilibrator provides standard free energy estimates for this calculation [15].

Enzyme kinetics set the forward capacity. Define the reference capacity as

$$V^\circ_{\max,j} \;=\; k^\circ_{\mathrm{cat},j}\, e^\circ, \tag{11}$$

where $k^\circ_{\mathrm{cat},j}$ is the turnover number and e° is a reference enzyme concentration. BRENDA reports turnover numbers [16], and BioNumbers reports cellular abundance scales [17]. The saturation factor gives the fraction of $V^\circ_{\max,j}$ available at the current substrate concentration. For a single-substrate enzyme, consider

$$E + S_j \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} [ES] \xrightarrow{k_{\mathrm{cat}}} E + P, \tag{12}$$

where $[ES]$ is the enzyme–substrate complex. The quasi-steady-state approximation and enzyme conservation, $[E]_T = [E] + [ES]$, give $[ES] = [E]_T[S_j]/(K_{M,j} + [S_j])$, where $K_{M,j} = (k_{-1} + k_{\mathrm{cat}})/k_1$. With $v_j = k_{\mathrm{cat}}[ES]$ and $V_{\max,j} = k_{\mathrm{cat}}[E]_T$, the Michaelis–Menten law gives [5]

$$v_j \;=\; V_{\max,j}\,\frac{[S_j]}{K_{M,j} + [S_j]} \qquad\Longrightarrow\qquad f_j([S_j]) \;=\; \frac{v_j}{V_{\max,j}} \;=\; \frac{[S_j]}{K_{M,j} + [S_j]} \in [0,1], \tag{13}$$

where $[S_j]$ is substrate concentration and $K_{M,j}$ is the Michaelis constant. The factor approaches zero at low substrate and one at saturation. Because f_j depends on $[S_j]$, the bound becomes state-dependent.

Gene expression sets enzyme abundance. Balances for transcript m_j and protein p_j are

$$\dot m_j = r_{X,j}\, u_j - (\theta_{m,j} + \mu)\, m_j + \lambda_j, \tag{14}$$

$$\dot p_j = r_{L,j}\, w_j\, m_j - (\theta_{p,j} + \mu)\, p_j, \tag{15}$$

where $r_{X,j}$ and $r_{L,j}$ are transcription and translation capacities, u_j and w_j are dimensionless controls, $\theta_{m,j}$ and $\theta_{p,j}$ are degradation constants, and λ_j is basal transcription. Growth adds μ to both loss rates. At steady state,

$$m^\star_j = \frac{r_{X,j} u_j + \lambda_j}{\theta_{m,j} + \mu}, \qquad p^\star_j = \frac{r_{L,j} w_j\, m^\star_j}{\theta_{p,j} + \mu}, \qquad (e/e^\circ) = \frac{p^\star_j}{e^\circ}, \tag{16}$$

Gene–protein–reaction rules map $p^\star_j$ to catalytic activity. These rules represent single enzymes, multisubunit complexes, and isozymes. Measured transcript or protein abundance can also set (e/e°) directly and produce condition-specific capacities.

Active and inactive configurations provide one statistical-mechanical description of allosteric, transcriptional, and translational control (Fig. 2). For an enzyme, promoter, or translation complex, the control is the statistical weight of the active set divided by the weight of all configurations. Let $\mathcal{S}_q$ be the configurations for process q, where q denotes catalysis, transcription, or translation. Configuration $s \in \mathcal{S}_q$ has energy ϵ_s and weight

$$\omega_s(\mathbf{x}) = g_s(\mathbf{x}) e^{-\beta\epsilon_s}, \qquad Z_q(\mathbf{x}) = \sum_{r\in\mathcal{S}_q} \omega_r(\mathbf{x}), \qquad p_s(\mathbf{x}) = \frac{\omega_s(\mathbf{x})}{Z_q(\mathbf{x})}, \tag{17}$$

where $\beta = 1/(k_B T)$ and Z_q is the partition function. The factor $g_s(\mathbf{x})$ includes ligand concentration, binding stoichiometry, and degeneracy. A state with n bound effector molecules, for example, has a concentration factor proportional to $(x/K)^n$.

Let $\mathcal{A}_q \subseteq \mathcal{S}_q$ contain the active configurations and let $\mathcal{I}_q = \mathcal{S}_q \setminus \mathcal{A}_q$ contain the inactive configurations. The control is the probability of the active set:

$$\Phi_q(\mathbf{x}) = \sum_{s\in\mathcal{A}_q} p_s(\mathbf{x}) = \frac{\displaystyle\sum_{s\in\mathcal{A}_q} g_s(\mathbf{x})e^{-\beta\epsilon_s}}{\displaystyle\sum_{r\in\mathcal{S}_q} g_r(\mathbf{x})e^{-\beta\epsilon_r}} = \frac{Z_{\mathcal{A}_q}}{Z_{\mathcal{A}_q} + Z_{\mathcal{I}_q}}. \tag{18}$$

Here $Z_{\mathcal{B}} = \sum_{s\in\mathcal{B}} \omega_s$ is the partition sum over subset $\mathcal{B}$. Active states appear in the numerator and denominator; inactive states appear only in the denominator. Therefore $0 \le \Phi_q \le 1$. An effector increases or decreases Φ_q according to the states it stabilizes. Choose configuration 0 as a reference and define $\Delta\epsilon_s = \epsilon_s - \epsilon_0$. Then

$$K_{\star,s} \equiv \frac{\omega_s/g_s}{\omega_0/g_0} = e^{-\beta\Delta\epsilon_s}, \qquad \frac{\omega_s}{\omega_0} = \frac{g_s(\mathbf{x})}{g_0(\mathbf{x})}K_{\star,s}. \tag{19}$$

With $\omega_0 = g_0 = 1$, the state weight is $\omega_s = g_s(\mathbf{x})K_{\star,s}$. A term such as $K_2 f_{TL}$ combines the energetic factor $K_2 = e^{-\beta\Delta\epsilon_2}$ with the available ligand-bound transcription factor f_{TL}. Fitted values of K_1 and K_2 may also include constant polymerase concentration, standard-state terms, or other fixed contributions. They need not be measured dissociation constants.

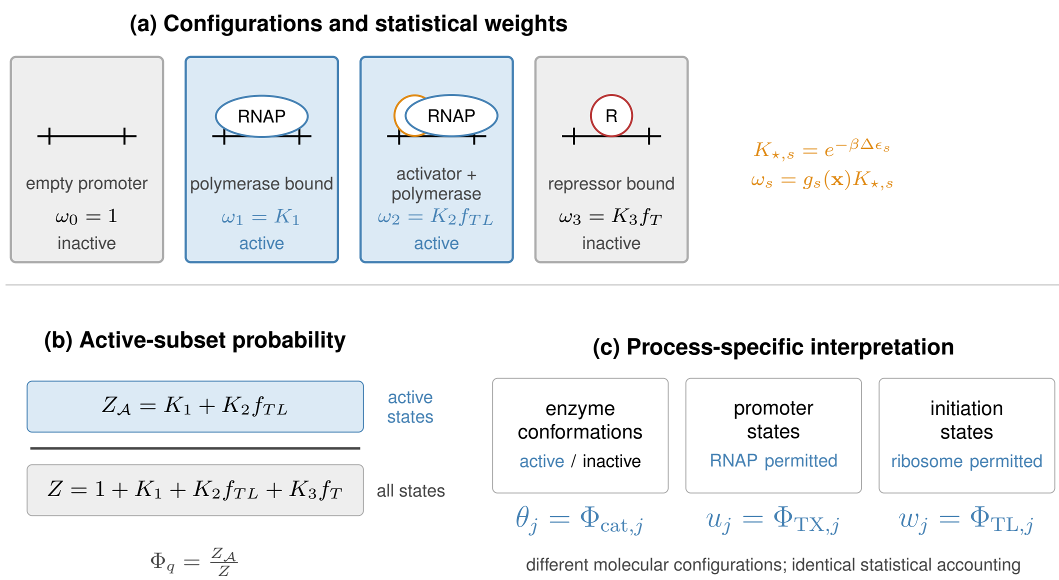

Figure 2: Active-subset construction of biological control factors. (a) Configuration weights combine relative energetics with regulator availability; blue states permit the process and gray states do not. (b) Active states enter the numerator, while all states enter the partition-function denominator. The promoter shown is schematic. (c) Specifying the relevant configurations gives allosteric activity θ_j, transcriptional control u_j, or translational control w_j from the same statistical accounting.

Moon, Voigt, and coworkers used this construction for inducible promoters [18]. The Lux promoter has weights 1, K_1, and $K_2 f_{TL}$ for the empty, polymerase-bound, and polymerase–LuxR-bound states. Both polymerase-bound states permit transcription. The Tac promoter has weights 1,

K_1, and $K_2 f_T$ for the empty, polymerase-bound, and repressor-bound states. Only the polymerase-bound state permits transcription. Normalization gives

$$u_{\mathrm{Lux}} = \frac{K_1 + K_2 f_{TL}}{1 + K_1 + K_2 f_{TL}}, \qquad u_{\mathrm{Tac}} = \frac{K_1}{1 + K_1 + K_2 f_T}. \tag{20}$$

Here f_{TL} is the ligand-bound transcription-factor fraction and f_T is the ligand-free fraction. Increasing inducer adds weight to an active Lux state and removes weight from an inactive Tac state. Both controls are special cases of Equation (18). A two-state model gives a familiar special case. If the inactive state has weight $\omega_I = 1$ and the effector-stabilized active state has weight $\omega_A = (x/K)^n e^{-\beta \Delta \epsilon}$, then

$$\Phi_q(x) = \frac{(x/K)^n e^{-\beta \Delta \epsilon}}{1 + (x/K)^n e^{-\beta \Delta \epsilon}}, \tag{21}$$

where $\Delta \epsilon$ is the active-state energy relative to the inactive state. If the effector instead stabilizes the inactive state, its weight appears only in the denominator and the control decreases with x.

Equation (18) gives each process-specific factor:

$$\theta_j = \Phi_{\mathrm{cat},j}, \qquad \bar{u}_j = \Phi_{\mathrm{TX},j}, \qquad w_j = \Phi_{\mathrm{TL},j}. \tag{22}$$

The factor θ_j multiplies catalytic capacity in Equation (9). The factor $\bar{u}_j$ controls transcription, and w_j controls translation in Equation (15). These controls are distinct from the degradation constants $\theta_{m,j}$ and $\theta_{p,j}$. Split the transcriptionally active set into regulated and basal states. Let $\mathcal{A}_{\mathrm{TX}}^{\mathrm{reg}}$ contain the regulated states and $\mathcal{A}_{\mathrm{TX}}^{\mathrm{basal}}$ contain states that support transcription without the regulated input. Then

$$\bar{u}_j = \frac{Z_{\mathcal{A}_{\mathrm{TX}}^{\mathrm{reg}}}}{Z_{\mathrm{TX}}} + \frac{Z_{\mathcal{A}_{\mathrm{TX}}^{\mathrm{basal}}}}{Z_{\mathrm{TX}}} = u_j + u_j^{\dagger}, \qquad \lambda_j \equiv r_{X,j} u_j^{\dagger}. \tag{23}$$

The regulated term u_j enters Equation (14). The basal term $u_j^{\dagger}$ defines λ_j. Moon, Voigt, and coworkers used this promoter-state construction [18]. Adhikari and coworkers extended it to transcriptional and translational control [19].

These factors require thermodynamic, kinetic, expression, and regulatory data. When data are unavailable, set the correction factors to one, $f_j \approx 1$, $(e/e^{\circ}) \approx 1$, and $\theta_j \approx 1$. This gives the baseline bound

$$-\delta_j V_{\max,j}^{\circ} \;\le\; \hat{v}_j \;\le\; V_{\max,j}^{\circ}. \tag{24}$$

Equation (24) retains only reversibility and reference capacity. Equation (9) shows which omitted factors require additional data. Each factor has a precedent in constraint-based modeling. Thermodynamics- based flux analysis uses free energies and metabolite activity ranges to exclude infeasible states [20]. Enzyme-constrained FBA uses turnover numbers and enzyme abundance; GECKO applies this approach at genome scale [21]. Metabolism-and-expression models couple transcription and translation to metabolism [22]. Regulatory FBA uses Boolean expression rules [23], whereas PROM uses data-derived interaction probabilities [24]. The allosteric factor θ_j represents direct regulation of enzyme activity. Equation (9) places these established terms in one notation. The balances above use aggregate transcription and translation capacities, $r_{X,j}$ and $r_{L,j}$. A sequence-resolved model can instead calculate nucleotide, amino-acid, and energy demands from each gene and protein sequence and add those reactions to $\mathbf{S}$ [9]. Expression then becomes part of the flux calculation. Vilkhovoy and coworkers used this construction for cell-free protein synthesis [25]; the later capstone returns to that model.

5 Worked Example: Urea-Cycle Metabolism

This example uses thermodynamic and kinetic data to set flux bounds for a urea-cycle network with four cycle reactions, a competing nitric oxide synthase reaction, and exchange with the environment (left panel, Fig. 3). The inputs come from public, cross-organism databases rather than HL-60-specific measurements. The model uses the steady-state constraint $\mathbf{S}\hat{\mathbf{v}} = \mathbf{0}$ from Equation (7) and the linear program in Equation (8). The application-specific quantities are $\mathbf{S}$, $\boldsymbol{\ell}$, $\mathbf{u}$, and $\mathbf{c}$. Argininosuccinate synthetase (v_1, EC 6.3.4.5) forms argininosuccinate. Argininosuccinate lyase (v_2, EC 4.3.2.1) produces arginine and fumarate. Arginase I (v_3, EC 3.5.3.1) produces ornithine and urea, and ornithine transcarbamylase (v_4, EC 2.1.3.3) regenerates citrulline. Nitric oxide synthase (v_5, EC 1.14.13.39) competes with arginase for arginine and produces nitric oxide and citrulline. The five enzymatic reactions involve eighteen metabolites. Fourteen metabolites cross the system boundary through exchange reactions b_1 through b_{14}. We use the secretion-positive convention $M_i \to \emptyset$: positive flux denotes secretion, and negative flux denotes uptake. Arginine, citrulline, ornithine, and argininosuccinate remain inside the network. The resulting matrix has eighteen metabolite balances and nineteen reactions, so $\mathbf{S} \in \mathbb{R}^{18\times 19}$.

We first set the thermodynamic factor in Equation (10). eQuilibrator gave the following standard reaction free energies [15]: $\Delta G^\circ_{v_1} = -4.3$, $\Delta G^\circ_{v_2} = -5.5$, $\Delta G^\circ_{v_3} = -51.0$, $\Delta G^\circ_{v_4} = -30.3$, and $\Delta G^\circ_{v_5} = -1220$ kJ/mol. The threshold ΔG^*_j of Equation (10) was set to -10 kJ/mol. This is a heuristic cutoff, not a value derived from a stated physiological concentration range. The rule makes v_1 and v_2 reversible and v_3, v_4, and v_5 irreversible. Exchange reactions retain wide bounds in both directions because their direction depends on the medium. We next set the kinetic capacities in Equation (11). Turnover numbers came from BRENDA [16]. Each was combined with the reference abundance $e^\circ = 0.01$ mmol/gDW, an illustrative intracellular enzyme concentration consistent with BioNumbers [17]. Argininosuccinate synthetase and nitric oxide synthase have $k^\circ_{\text{cat}} = 10.0\ \text{s}^{-1}$, which gives $V^\circ_{\max,v_1} = V^\circ_{\max,v_5} = 360\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$; arginase I has $k^\circ_{\text{cat}} = 190.0$ s^{-1}, giving $V^\circ_{\max,v_3} = 6840\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$; and ornithine transcarbamylase has $k^\circ_{\text{cat}} = 410.0$ s^{-1}, giving $V^\circ_{\max,v_4} = 14760\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$. Argininosuccinate lyase has the lowest turnover number, $k^\circ_{\text{cat},v_2} = 3.28\ \text{s}^{-1}$, which gives $V^\circ_{\max,v_2} = 118.08\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$, the smallest of the five capacities. Exchange reactions retain wide capacity bounds.

The model has no substrate, expression, or regulatory data, so setting $f_j = (e/e^\circ) = \theta_j = 1$ yields the reported optimal flux distribution (right panel, Fig. 3). The enzymatic bounds therefore reduce to $-\delta_j V^\circ_{\max,j} \le \hat{v}_j \le V^\circ_{\max,j}$ for v_1 through v_5. The fourteen exchange reactions use wide bounds of $-1000 \le \hat{v}_j \le 1000\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$. The objective is urea export. Because b_4 is written as urea $\to \emptyset$, we set $c_{b_4} = 1$ and all other entries of $\mathbf{c}$ to zero. The linear program [10] gave a unique optimum. Reactions v_1 through v_4 each carried $118.08\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$, the capacity of argininosuccinate lyase. The competing nitric oxide synthase branch v_5 carried zero flux because the objective assigns all available arginine to urea production. The optimal urea export was $118.08\ \text{mmol}\,\text{gDW}^{-1}\,\text{h}^{-1}$, attained at $\hat{v}_{b_4} = 118.08$, and every exchange flux carrying carbon or nitrogen into or out of the cycle, carbamoyl phosphate and aspartate on the uptake side, urea and fumarate on the secretion side, scaled to match it. Flux-variability analysis gave the same minimum and maximum for every reaction when the objective was held at its optimum. The optimal flux distribution is therefore unique to numerical tolerance.

The Monte Carlo analysis treats the database inputs as uncertain rather than exact and compares the effect of imputing the missing saturation factor (Fig. 4). We assigned lognormal uncertainty of about a factor of two to each turnover number and the reference abundance, and normal uncertainty of a few kilojoules per mole to each standard free energy. For four reactions, we also sampled substrate concentrations and Michaelis constants from Park and coworkers [26]. Argininosuccinate lyase lacked a measured substrate concentration, so its saturation factor remained at one. We then solved the linear program for ten thousand parameter draws using Algorithm 1. This Monte Carlo calculation produced a right-skewed urea-export distribution with median 105 and mean 156 $\mathrm{mmol\,gDW^{-1}\,h^{-1}}$ and a central ninety-five percent interval of 17 to 614. Varying only capacity and free-energy inputs gave a median of 113 $\mathrm{mmol\,gDW^{-1}\,h^{-1}}$. Adding the four measured saturation factors changed the median to 105, so those factors had little effect. Most of the uncertainty came from the sampled capacity of argininosuccinate lyase. The reversibility switches did not bind because the optimal cycle flux was forward, and the other four enzymes usually retained excess capacity. The nitric oxide synthase flux remained small overall. The missing saturation factor for argininosuccinate lyase matters more. When we imputed it from a physiological argininosuccinate concentration and a BRENDA Michaelis constant, the median fell to 50 $\mathrm{mmol\,gDW^{-1}\,h^{-1}}$ and the distribution widened toward zero. Better data for this limiting reaction would reduce the uncertainty most directly. This example used no HL-60-specific concentration, expression, or regulatory measurements. eQuilibrator set reaction direction, BRENDA set capacity scales, and the other factors remained at one. The result is limited mainly by v_2, while the competing v_5 branch carries little flux. The next example adds allosteric and expression control to a reaction bound.

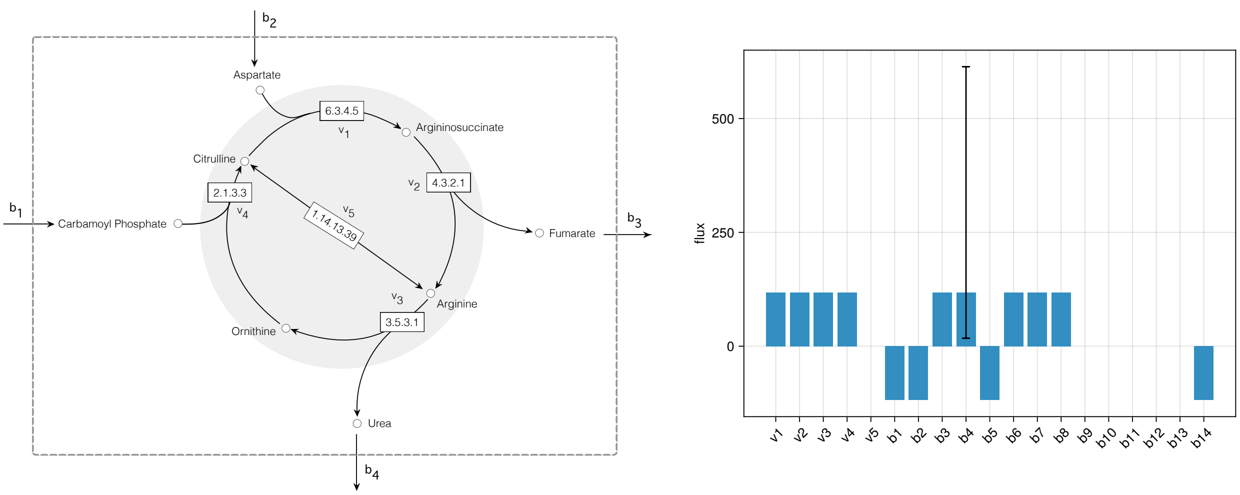

Figure 3: Urea-cycle network and optimal fluxes. Left: reactions v_1 through v_4 form the cycle, and v_5 is the competing nitric oxide synthase branch. Exchange reactions cross the dashed boundary; arginine, citrulline, ornithine, and argininosuccinate remain internal. Right: the optimal flux distribution for Equation (8). Positive exchange flux denotes secretion, and negative exchange flux denotes uptake. The cycle reactions carry 118.08 $\mathrm{mmol\,gDW^{-1}\,h^{-1}}$, the capacity of argininosuccinate lyase (v_2), while v_5 carries zero flux. The whisker on urea export b_4 shows the 2.5–97.5 percentile interval from the Monte Carlo calculation in Algorithm 1.

6 Worked Example: A Feedback-Inhibited Linear Pathway

The urea-cycle example used thermodynamic and kinetic factors but set the expression ratio (e/e°) and allosteric factor θ_j to one. This example includes both controls. The end product inhibits the existing enzyme and represses transcription of its gene. These mechanisms change enzyme activity and enzyme abundance, respectively, and they act on different timescales. We represent them in a linear pathway with one controlled step:

$$\emptyset \xrightarrow{r_0} X_1 \xrightarrow{r_1} X_2 \xrightarrow{r_2} X_3 \xrightarrow{r_3} \emptyset, \tag{25}$$

Algorithm 1 Monte Carlo propagation of urea-cycle parameter uncertainty

Require: nominal k°_{cat}, e°, ΔG°; nominal saturation inputs $\overline{[S_j]}, \overline{K_{M,j}}$ where measured; spreads $\sigma_{\ln}, \sigma_{\Delta G}$; threshold ΔG^*; draws N; seed

seed the random generator
for $i = 1$ to N **do**
    $e \leftarrow e^{\circ} \exp(\sigma_{\ln} Z_e)$ ▷ one shared abundance draw, $Z_e \sim \mathcal{N}(0, 1)$
    for each enzymatic reaction j **do**
        $k_{\mathrm{cat},j} \leftarrow k^{\circ}_{\mathrm{cat},j} \exp(\sigma_{\ln} Z_{k,j})$, $\quad \Delta G_j \leftarrow \Delta G^{\circ}_j + \sigma_{\Delta G} Z_{\Delta G,j}$ ▷ independent $Z_{k,j}, Z_{\Delta G,j} \sim \mathcal{N}(0, 1)$ per reaction
        $\delta_j \leftarrow \mathbb{1}[\Delta G_j > \Delta G^*]$
        if substrate data measured for reaction j **then**
            $[S_j] \leftarrow \overline{[S_j]} \exp(\sigma_{\ln} Z_{S,j})$, $\quad K_{M,j} \leftarrow \overline{K_{M,j}} \exp(\sigma_{\ln} Z_{K,j})$ ▷ independent per reaction
            $f_j \leftarrow [S_j]/(K_{M,j} + [S_j])$
        else
            $f_j \leftarrow 1$
        end if
        $V_{\max,j} \leftarrow 3600\, k_{\mathrm{cat},j}\, e$; $\quad u_j \leftarrow V_{\max,j} f_j$, $\quad \ell_j \leftarrow -\delta_j V_{\max,j} f_j$
    end for
    $\hat{\mathbf{v}}^{(i)} \leftarrow \arg\max\{\mathbf{c}^{\top}\mathbf{v} : \mathbf{S}\mathbf{v} = \mathbf{0},\ \boldsymbol{\ell} \le \mathbf{v} \le \mathbf{u}\}$
end for
return per-reaction mean, standard deviation, and 2.5/50/97.5 percentiles of $\{\hat{\mathbf{v}}^{(i)}\}_{i=1}^{N}$

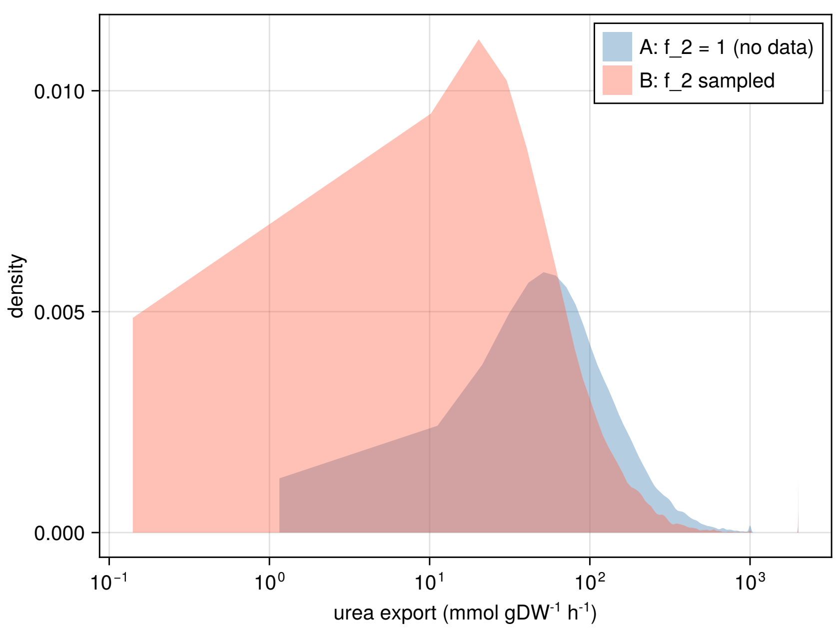

Figure 4: Sensitivity of urea export to saturation information. Configuration A (blue) samples four saturation factors derived from data reported by Park and coworkers [26] and holds $f_2 = 1$ at the argininosuccinate lyase bottleneck, whose substrate is unmeasured, giving a median urea export of 105 mmol gDW^{-1} h^{-1}. Configuration B (red) imputes f_2 from a physiological argininosuccinate concentration and a BRENDA Michaelis constant and samples it as well, shifting the median to 50 mmol gDW^{-1} h^{-1} and widening the band toward zero. The horizontal axis is logarithmic.

where r_0 is the committed uptake step, r_1 and r_2 are interconversions, and r_3 exports X_3. The end product X_3 controls both the activity and abundance of the enzyme for r_0. We generated a kinetic reference with a Biochemical Systems Theory (BST) S-system [6, 7]. The model includes five dynamic states: metabolites X_1, X_2, and X_3, transcript m, and enzyme E_0. BSTModelKit.jl was used to integrate these states to steady state [8]. The end product enters the committed-step rate with kinetic order $-a$ and the transcription rate with kinetic order $-b$. Increasing X_3 therefore lowers both enzyme activity and enzyme production. Metabolic fluxes obey $\mathbf{S}\hat{\mathbf{v}} = \mathbf{0}$, while transcript and protein balances include degradation and growth dilution. The committed-step rate is

$$\begin{aligned} v_{r_0} &= V^{\circ}_{\max,r_0}\,(e/e^{\circ})\,\theta_{\mathrm{BST}}, \\ \theta_{\mathrm{BST}} &= (X_3/X_3^{\circ})^{-a}, \qquad a = 0.4, \\ \text{transcription} &\propto (X_3/X_3^{\circ})^{-b}, \qquad b = 0.6, \end{aligned} \tag{26}$$

where $X_3^{\circ} = 1$ in the concentration units used by the simulation. The normalized ratios are dimensionless.

The kinetic reference combines expression timescales with metabolic feedback and gives a steady-state throughput $T^{\star} \approx 2.66$ (Fig. 5). It uses a 330-residue enzyme, a 990-nucleotide gene, polymerase elongation at 60 nucleotides per second, and ribosome elongation at 16 amino acids per second. These values provide illustrative expression timescales. The steady-state abundance also depends on an mRNA half-life of 2.5 minutes, a protein half-life of 35 minutes, and a culture doubling time of 60 minutes. The same growth rate μ appears in both balances, but its relative contribution differs because transcript and protein have different degradation rates. Growth accounts for about four percent of transcript clearance, $\theta_m + \mu \approx 0.289\ \mathrm{min}^{-1}$, and about thirty-seven percent of enzyme clearance, $\theta_p + \mu \approx 0.031\ \mathrm{min}^{-1}$. At steady state, every reaction in the linear chain carries the same flux T. Because export is first order in X_3, $X_3 = T/k_3$. Substitution of the activity and expression controls gives one consistency condition for X_3. The numerical solution was $X_3^{\star} \approx 4$, with activity factor $\theta^{\star}_{\mathrm{BST}} \approx 0.574$, an expression ratio $(e/e^{\circ})^{\star} \approx 0.464$. Basal transcription λ, defined by Equations (14) and (23), keeps expression above zero. The values $a = 0.4$ and $b = 0.6$ make both controls affect the result; neither control alone produces the reference throughput.

The FBA comparison keeps the chain's stoichiometry unchanged and places the entire dual-control content on the committed-step bound (Fig. 5). The bound is

$$0 \;\le\; \hat{v}_{r_0} \;\le\; V^{\circ}_{\max,r_0}\,(e/e^{\circ})\,\theta_{\mathrm{FBA}}(X_3), \qquad V^{\circ}_{\max,r_0} = 10, \tag{27}$$

which is the expression-and-regulation form of Equation (9). The thermodynamic and saturation factors remain at one. To avoid copying the BST rate law into the FBA bound, we represent the activity factor with a bounded two-state Hill function:

$$\theta_{\mathrm{FBA}}(X_3) \;=\; \frac{K^n}{K^n + X_3^n}, \qquad K = 5, \quad n = 2, \tag{28}$$

the repressive form of Equation (21). The active state has unit weight, and the inactive state has weight $(X_3/K)^n$. The parameters K and n were specified independently of the BST order a. Both factors were evaluated from the steady-state enzyme abundance and X_3, not from the reference flux. Four FBA solutions used the same stoichiometric matrix and differed only in the bound on r_0. With both factors set to one, the predicted flux was 10. Expression control alone reduced it to 4.64, activity control alone reduced it to 6.10, and both controls reduced it to 2.83. The dual-control result is within about six percent of the BST reference value, 2.66.

The parameter sweep addresses uncertainty in the Hill parameters, which were specified as $K = 5$ and $n = 2$ rather than fitted (Fig. 6). We therefore varied K from 3 to 8 and n from

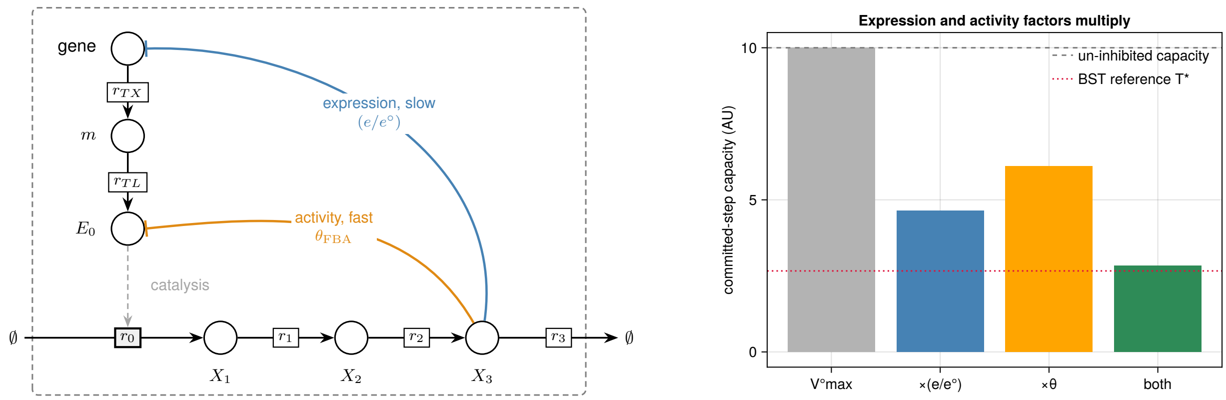

Figure 5: Dual-control feedback example. Left: the pathway and the transcription and translation steps that produce enzyme E_0. The end product X_3 represses enzyme expression and inhibits enzyme activity. Right: predicted r_0 flux with neither control, each control alone, and both controls. Expression control gives 4.64, activity control gives 6.10, and both give 2.83. The BST reference is $T^{\star} \approx 2.66$, and the dashed line marks the uninhibited capacity of 10.

1 to 3, recomputed θ_{FBA} at the reference X_3, and solved the program at each grid point. With both controls, the predicted flux ranged from about 1.4 to 4.1 and included the BST value of 2.66. The nominal parameters gave 2.83. Activity control alone kept the flux above about 2.97, while expression control alone kept it at 4.64. Thus, neither single-control model included the BST value over the tested parameter range, but the dual-control model did.

7 Integration in Practice

The worked examples used small networks with one degree of freedom, $n - m = 1$. This section considers two larger applications. The first combines metabolism, expression, and regulation in a dynamic cell-free model. The second changes reaction bounds to design a glycan-producing strain. Vilkhovoy and coworkers built a sequence-specific model of *E. coli* cell-free protein synthesis [25]. Transcription and translation reactions derived from DNA and mRNA sequences appear in the stoichiometric matrix with metabolism. Promoter and ribosome control functions u_j and w_j follow the biophysical models of Moon, Voigt, Adhikari, and coworkers [18, 19]. The dynamic model updates thermodynamic, kinetic, expression, and regulatory bounds during the reaction [12]. It was compared with time courses for 63 metabolites, mRNA, protein, and enzyme activity. Cell-free systems also expose accumulation directly. They have no growth dilution, so $\mu = 0$, but their molecular pools change during a batch. Therefore, the steady-state reduction to Equation (7) does not apply. Time-course measurements can instead evaluate the accumulation term in Equation (6).

Wayman and coworkers used a constraint-based model for strain design. They added the N-linked glycosylation pathway of *Campylobacter jejuni* onto a genome-scale reconstruction of *E. coli* metabolism and searched for gene deletions that couple designer-glycan production to growth [13]. Each candidate knockout sets a reaction bound to zero, after which the linear program is solved again. Enzyme attenuation or overexpression can instead decrease or increase the (e/e°) factor in Equation (9). The single Δgnd deletion, drawn from the model-identified strain families, increased measured glycan production nearly threefold relative to wild type. The model identified candidate perturbations; it did not predict the magnitude of this experimental increase. The two applications

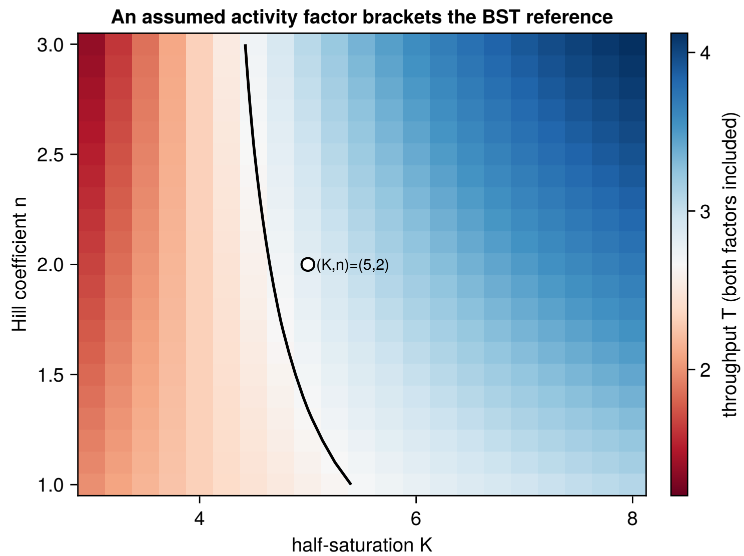

Figure 6: Sensitivity to the assumed activity-factor shape. The plot shows the dual-control flux while K and n vary in $\theta_{\text{FBA}} = K^n/(K^n + X_3^n)$. The color scale is centered on the BST reference $T^\star \approx 2.66$, and the black contour marks equal FBA and BST fluxes. The predicted flux ranges from about 1.4 to 4.1. The marked nominal point, $(K, n) = (5, 2)$, gives 2.83.

use bounds for different purposes. The cell-free model uses measurements to estimate fluxes. The glycan model changes bounds to test possible interventions. Strain design also requires a discrete search over deletions or expression changes, which can be formulated as a bilevel program [14].

8 Outlook

The worked examples used standard free energies, turnover numbers, and regulatory models to set the factors in Equation (9). Time-course data can instead be used to estimate turnover numbers, saturation constants, allosteric factors θ_j, and expression controls u_j and w_j. Tabulated values remain useful as prior information when dynamic data are unavailable. The same approach can be applied reaction by reaction in genome-scale models. These models have many flux distributions that satisfy $\mathbf{S}\hat{\mathbf{v}} = \mathbf{0}$, so condition-specific bounds are especially important. Setting an unknown factor to one preserves feasibility but may leave many fluxes unresolved. Time-course data also allow models to retain accumulation and growth dilution. Cell-free systems have $\mu = 0$ but can have substantial accumulation. Growing cell cultures can have both terms. When measurements resolve these changes, fluxes should be estimated from Equation (6) instead of imposing Equation (7).

Code and data availability

Source code, generated data, and instructions for reproducing the worked examples and figures are available at `https://github.com/varnerlab/MRW-BTC4-Chapter-Varner`. The repository includes the Julia project and manifest files needed to reproduce the computational environment.

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